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P.M.B 012, O. A. U., Ile-Ife, Osun State, Nigeria Tel.: +2347034584645 Email: dg.ceo@nacetem.gov.ng



TOWARD INCLUSIVE AGRICULTURAL INNOVATIONS: RETHINKING TEAM COMPOSITION IN NIGERIAN RESEARCH INSTITUTES

Introduction

Agricultural R&D is vital to boosting food production, improving resilience to climate change, and enhancing Nigeria's food security. In this context, Federal Government-funded research institutions play a critical role. However, the effectiveness of these agricultural research institutions is hinged on project funding, the quality and composition of project research teams. The composition of R&D teams—including the gender distribution, academic qualifications, specialisation, and rank of members—affects research output and innovation. For instance, diversity in project team configuration generates more innovative solutions. Furthermore, ensuring that R&D project Principal Investigators (PIs) and other team members possess relevant qualifications and specialisations is essential for the success of agricultural R&D projects.

Consequently, this brief provides empirical insights into the dynamics of R&D project teams in Nigeria's agricultural research institutions, identifying critical gaps and opportunities for improvement.

Method

The methodology involved extracting project details (titles and funding allocations) from National Annual Budgetary Reports, organising them by institute, and analysing decade-long patterns to distinguish original initiatives from repeated efforts. This comprehensive approach ensured that the study evaluated only distinct, non-redundant projects that contributed meaningfully to agricultural innovation. The population for this study comprises R&D projects conducted across Nigeria's 17 federal agricultural research institutes over 10 years (2013-2022). We compiled data on 828 R&D projects from budget reports spanning 2013 to 2022. Through rigorous screening, we identified and eliminated duplicate project titles to isolate 142 unique crop production (including horticulture) projects across 9 research institutes. Finally, 50 agricultural R&D projects were selected from the 2018–2022 period to capture recent R&D trends.

KEY FINDINGS

Profile of Principal Investigators (PIs)

The gender distribution (65% male and 35% female) is clearly indicative of male dominance in the composition of R&D project team members, thus suggesting gender imbalance and disparity in the leadership (Fig 1).

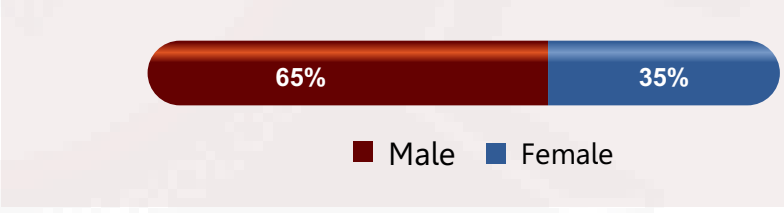


Fig. 1: Principal Investigators of R&D Project by Gender

The majority of project members (90 %) hold doctoral degrees, indicating a highly educated and strong research capacity, which is critical for research projects aimed at addressing the complexities of food insecurity challenges in Nigeria (Smithy et al., 2021; Verma et al., 2024) (Fig 2).

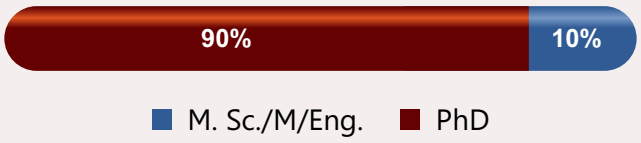


Fig. 2: Principal Investigators of R&D Project by Highest Academic Qualification

The most common ranks among project team leads are Professor/Director of Research (41%), Assistant Director of Research (23.1%), Senior Lecturer/Chief Research Officer (28.2%), and Lecturer I/Principal Research Officer (7.7%) (Figure 3). Figure 4 further illustrates the areas of specialisation of the Principal Investigators, with the majority having academic backgrounds in genetics, crop science, or plant breeding

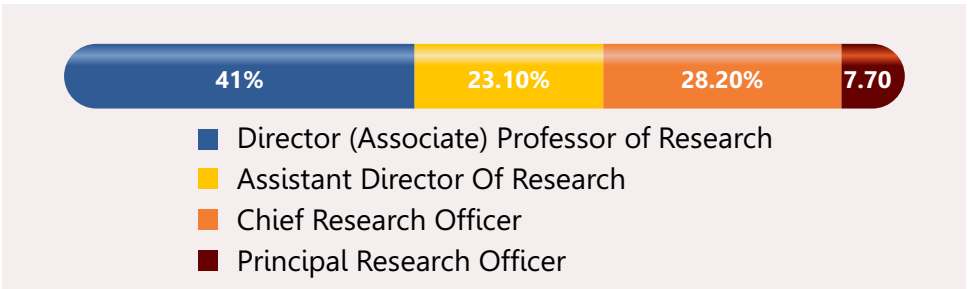


Fig. 3: R&D Project PIs by Rank

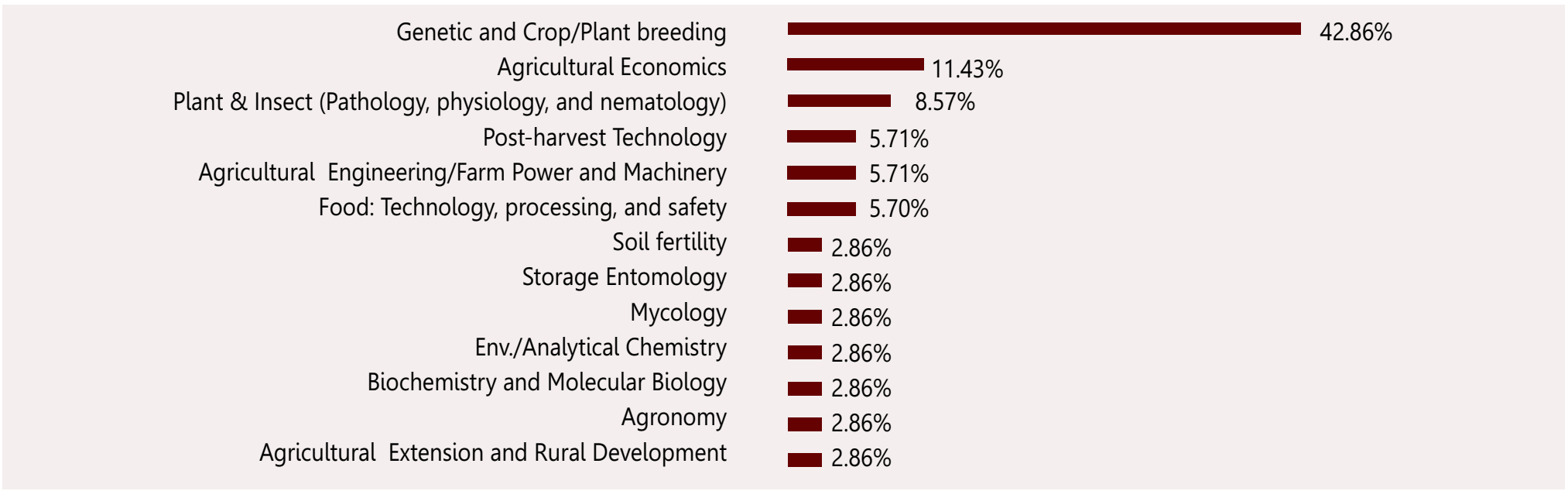


Fig. 4: PIs of R&D Project by Core Area of Specialization

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Profile of Other R&D Project Team Members (Excluding PIs)

Figure 5 shows a gender distribution of 65% female and 35% male, indicating that female researchers constituted the majority of the project team.

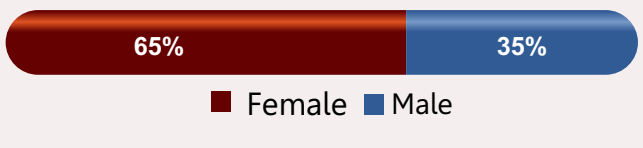


Fig. 5: Project Team Composition by Gender

Figure 6 presents the academic qualifications of the project team members: 53% hold PhDs, 31% have MSc degrees, and 16% possess BSc degrees. The high proportion of postgraduate degree holders (84%) is notable and suggests strong potential to enhance the depth and quality of the research outputs and outcomes.

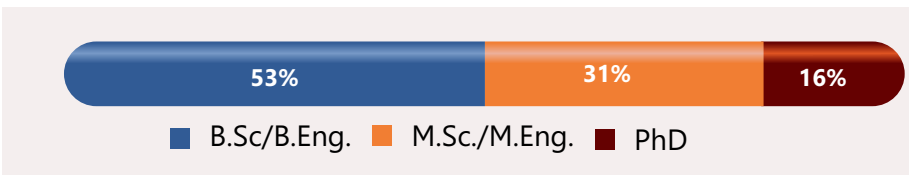


Fig. 6: Project Team Composition by Academic Qualification

Figure 7 shows that 28% of the team members were Professors or Directors of Research, 20% were Senior Lecturers or Chief Research Officers, 18% were Lecturer I or Principal Research Officers, 15% were Lecturer II or Senior Research Officers, and 19% were non-academic staff such as laboratory technicians. This distribution reflects a strong involvement of experienced researchers in the implementation of research projects.

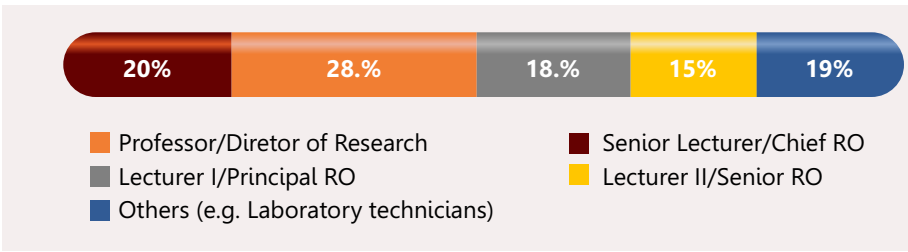


Fig. 7: Rank of Project Team members

Implications of the Key Findings

The findings highlight critical aspects of research team composition affecting agricultural R&D innovation in Nigeria. A gender imbalance (male or female-dominated) may limit diverse innovative perspectives in research projects, thus requiring policies to ensure equitable participation in agricultural research projects. The high percentage of PhD holders (90% for PIs or 53% for other project team members) suggests strong expertise. Nonetheless, over-reliance on senior researchers (41% Professors/Directors) could hinder fresh ideas, calling for greater inclusion of mid-level and junior researchers. Mixed qualifications (PhD, MSc, BSc) could enhance knowledge transfer, but also require structured roles to maximise contributions. In conclusion, to drive agricultural innovative research, it is important to balance hierarchy in research team composition (reducing overrepresentation of senior ranks), promote gender diversity, and leverage varied qualifications. Addressing these gaps can optimise team dynamics, fostering more adaptive and impactful agricultural R&D outcomes in Nigeria.

Conclusion

The growing presence of highly qualified personnel in the surveyed Nigerian agricultural research institutes is a promising trend that should be sustained and replicated across other R&D sectors. Harnessing the expertise of individuals with advanced degrees can greatly improve the quality, relevance, and impact of research outputs. Equally crucial is the promotion of gender inclusivity within the research landscape—women should be actively encouraged to join research teams and supported to take on leadership roles, including that of Principal Investigator. Promoting both academic excellence and gender diversity will enhance the effectiveness of R&D institutions in addressing Nigeria's agricultural and broader development challenges

Contributors: Ayo-Lawal R.A., Atoyebi M.K., Ogungbemi A.A., Omimakinde E.A., Okpuru O. A., and Aodu B.A.